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ASSESSMENT 3

CSA1006-SOFTWARE ENGINEERING

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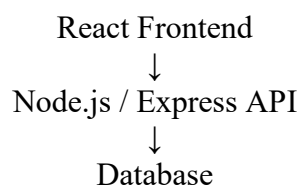
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Problem 1: Full-Stack Web App Development for an E-Commerce Dashboard

Scenario

A development team is building an e-commerce analytics dashboard using React for the frontend, Node.js and Express for the backend, and a database to store information. The application displays real-time data through APIs. The dashboard should provide interactive charts, fast loading speed, and a user-friendly interface.

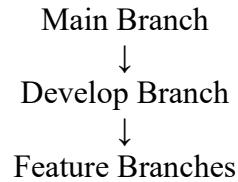
System Architecture



Explanation

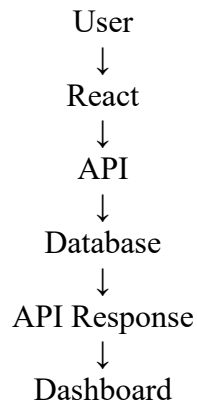
The React frontend sends requests to the API. The backend processes the request and retrieves data from the database. The API returns JSON data, which is displayed on the dashboard.

Git Branching Strategy



Feature developers work independently and create pull requests before merging into the develop branch.

API Data Flow



Pull Request Checklist

- Code reviewed
- Tests passed
- No errors
- Documentation updated
- Approved before merge

Problem 2: Dockerizing a News Aggregator Application

Scenario

The backend of a news aggregator application runs inside a Docker container to ensure the application behaves the same on all systems.

Dockerfile

- Use Node.js base image
- Copy project files
- Install dependencies
- Expose application port
- Run application

Build Command

`docker build -t news-app .`

Run Command

`docker run -p 3000:3000 news-app`

Push Image to Docker Hub

1. Login
2. Tag Image
3. Push Image

Advantages of Docker

- Same environment
- Easy deployment
- Better portability
- Faster testing
- Easy maintenance

Problem 3: Kubernetes Deployment

Scenario

A video streaming application requires automatic scaling and high availability.

Kubernetes Deployment

Replica Count: 3

Container: Video API

Port: 8080

Service

LoadBalancer

Horizontal Pod Autoscaler

Automatically increases pods when CPU usage becomes high.

Helm Structure

charts/

templates/

values.yaml

Chart.yaml

Benefits

- High availability
- Automatic scaling
- Better resource utilization

Problem 4: Git Collaboration

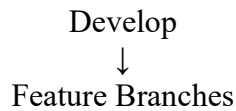
Scenario

Developers are pushing directly into the main branch, creating conflicts.

Workflow

Main





Every feature is reviewed before merging.

Pull Request Rules

- At least one reviewer
- Unit tests passed
- Linting passed
- No merge conflicts

Merge Conflict Resolution

1. Pull latest changes.
2. Resolve conflicts.
3. Test again.
4. Commit.
5. Push.

Problem 5: High-Performance Algorithm Optimization

Scenario

A ride-sharing platform needs to match riders with drivers quickly during peak hours.

Bottlenecks

- Slow search
- High response time
- Large number of requests

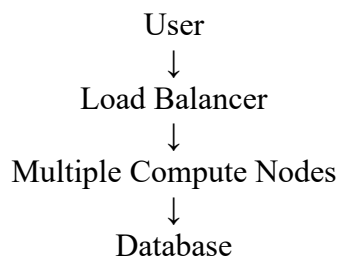
Optimized Solution

- Better data structures
- Caching
- Parallel processing

Concurrency Model

- Threads
- Async Programming
- Worker Processes

Distributed Workflow



Benefits

- Faster matching
- Better scalability

- Reduced delay

CONCLUSION

The proposed solutions improve application performance, scalability, security, and maintainability. Technologies such as React, Docker, Kubernetes, Git, and optimized algorithms provide efficient software development and deployment. Following these approaches ensures reliable and high-quality software systems.

REFERENCES :

- React Team. (2025). *React Documentation*. Available: <https://react.dev>
- Node.js Foundation. (2025). *Node.js Documentation*. Available: <https://nodejs.org/docs>
- Docker Inc. (2025). *Docker Documentation*. Available: <https://docs.docker.com>
- Kubernetes Authors. (2025). *Kubernetes Documentation*. Available: <https://kubernetes.io/docs>
- GitHub Inc. (2025). *GitHub Documentation*. Available: <https://docs.github.com>
- Deng, S., Zhao, H., Huang, B., Zhang, C., Chen, F., Deng, Y., Yin, J., Dustdar, S., & Zomaya, A. (2024). **Cloud-Native Computing: A Survey From the Perspective of Services**. *Proceedings of the IEEE*. DOI: 10.1109/JPROC.2024.3353855.
- Shrestha, R., & Ray, A. (2024). **Streamlining Application Deployment: A CI/CD Pipeline for Kubernetes**. *IEEE International Conference on Cloud Engineering (IC2E 2024)*. DOI: 10.1109/IC2E61754.2024.00038.